Sources Sought Synopsis (SSS)

For P26-005 Desiccant Machine

Tinker Air Force Base, OK

Ref. # P26-005

26 February 2026

Desiccant Machine

CAUTION:

This market survey is being conducted to identify potential sources that may possess the **expertise**, **capabilities**, and **experience** to meet the requirements for qualification and production of the Desiccant Machine for use by United States Air Force (USAF) personnel. The level of security clearance and amount of foreign participation in this requirement has not been determined. Contractors/Institutions responding to this market research are placed on notice that participation in this survey may not ensure participation in future solicitations or contract awards. The government will not reimburse participants for any expenses associated with their participation in this survey.

INSTRUCTIONS:

1. Below is a document containing a description of the Desiccant Machine requirement and a Contractor Capability Survey, which allows you to provide your company's capability.
   a) SOW
   b) Item Description

2. If, after reviewing these documents, you desire to participate in the market research, you should provide documentation that supports your company's capability in meeting these requirements. Failure to provide documentation may result in the government being unable to adequately assess your capabilities. If you lack sufficient experience in a particular area, please provide details explaining how you would overcome the lack of experience/capabilities in order to perform that portion of the requirement (i.e., teaming, subcontracting, etc.)
   a. Identify any areas of work that your organization believes should be broken out exclusively for Small Business.

3. Both large and small businesses are encouraged to participate in this Market Research. Joint ventures or teaming arrangements are encouraged.

4. Questions relative to this market survey should be addressed to Tori Wallace, 405-734-0453.

Desiccant Machine

PURPOSE/DESCRIPTION

The Government is conducting market research to identify potential sources that possess the production data/repair data, expertise, capabilities, and experience to meet qualification requirements for the Desiccant Machine. The following descriptions/specifications are from the Item Description.

The Contractor shall be responsible for providing us with the following:

Vacuum Test Engine

- Connect apparatus to [redacted] engine and adjust pressure to 27.7 +/- 0.3 inches of H_2O.
- Maintain pressure for 5 seconds before recording the flow rate.
- Shall be able to detect flow rate as little $\leq 0 \pm 0.010$ scfm.

Pressure Test Engine

- Connect apparatus to [redacted] engine and adjust pressure to 27.7 +/- 0.3 inches of H_2O.
- Maintain pressure for 5 seconds before recording the flow rate.
- Shall be able to detect flow rate as little as 0.045 scfm.

Size of Engine

Engine Length: 49.35”, Width: 15.56”, Height: 23.61”

System Performance Requirements

- System shall be capable of vacuum and pressure calibration using internal regulators and indicator gauges.

Calibration and Communication

- The contractor shall provide calibration instruction and allow calibration of the system using in house apparatus (PMEL).
- The system shall include ports for data acquisition and communication, enabling monitoring of flow and pressure during test operations.
- Flow and pressure gauges must be accurate to detect flow rate and is clearly readable.

Safety and Electrical

- The entire assembly shall operate on standard 110V AC power and include a grounded plug.
- The system shall include proper grounding mechanisms to prevent static discharge during operation.
- Emergency shutoff functionality shall be included.

Control and Regulation Requirements

- The system shall include clearly labeled and accessible valves/ switches to control including but not limited to
    - Vacuum Pump
    - Shop Air
    - Vacuum Regulator
    - Pressure Regulator
    - Mode Selector Valve
    - Instrument Power
- Control logic must support Mode 1 (vacuum) and Mode 2 (Pressure) operations.
- The operator interface shall be simple and intuitive, providing no more than two selectable options: Vacuum Test or Pressure Test.
- If implemented as a software interface, the selection process shall be limited to two inputs only (e.g. two buttons, two menu options, or toggle), with no unnecessary sub menus or advanced settings.
- The system shall include a fail safe default state that ensures the system powers on in a neutral/ off condition until one of the two modes is selected.

Document and Deliverables

- Contractor shall deliver:
    - Operator Manual
    - Calibration Procedure
    - Wiring and Pneumatic Diagram
    - Training Material
    - **Contractor shall also perform an onside demonstration and functional check with government personnel upon delivery to meet all requirements of the Statement of Work (SOW).**

Data Acquisition and Communication

- The current setup has two connections, "SWAGELOK PAT D-QF 4-B-316" on the inlet cover and a "9/16-18 thread" on the engine service island.
- The new apparatus connections shall include AN4- hose connections with stainless braided hosing.

- The connections on the hosing shall be SWAGELOK QC-4 quick connect fittings, featuring KEY 1 (as one hose for pressure) to SWAGELOK SS-QF4-B-4PF connection (the engine side) and Key 2 (second hose for vacuum), with the other end being AN6 to AN4 reducer male to male.
- All hosing shall be a minimum of 5 ft in length.
- To connect apparatus to base airline a 1/4 NPT (male) quick connect is necessary.

3. **Contractor shall provide delivery, installation, and performance verification for the Desiccant Machine IAW the Statement of Work (SOW).**

CONTRACTOR CAPABILITY SURVEY

Desiccant Machine

Part I. Business Information

Please provide the following business information for your company/institution and for any teaming or joint venture partners:

- Company/Institute Name:
- Address:
- Point of Contact:
- CAGE Code:
- Phone Number:
- E-mail Address:
- Web Page URL:
- Size of business pursuant to North American Industry Classification System (NAICS) Code: **333517**
  Based on the above NAICS Code, state whether your company is:
  - Small Business (Yes / No)
  - Woman Owned Small Business (Yes / No)
  - Small Disadvantaged Business (Yes / No)
  - 8(a) Certified (Yes / No)
  - HUBZone Certified (Yes / No)
  - Veteran Owned Small Business (Yes / No)
  - Service Disabled Veteran Small Business (Yes / No)
- Central Contractor Registration (CCR). (Yes / No)
- A statement as to whether your company is domestically or foreign owned (if foreign, please indicate the country of ownership).

Written responses, no facsimiles or e-mails please, must be received no later than close of business 14 April 2026. Please email two ***(2)*** your response to:

Desiccant Machine
Attn: Tori Wallace – Program Manager
Email: tori.wallace@us.af.mil

Questions relative to this market research should be addressed to Tori Wallace, (405) 590-7260.

Part II. Capability Survey Questions

A. General Capability Questions:

1. Describe briefly the capabilities of your facility and the nature of the goods and/or services you provide. Include a description of your staff composition and management structure.

2. Describe your company's past experience on previous projects similar in complexity to this requirement. Include contract numbers, a brief description of the work performed, period of performance, agency/organization supported, and individual point of contact (Contracting Officer or Program Manager).

3. What quality assurance processes and test qualification practices does your company employ? Please provide a description of your quality program (ISO 9001, AS9100, etc.)

4. Describe your company's capabilities for generating, handling, processing and storing classified material and data.

5. Describe your company's capabilities and experience in generating technical data, engineering drawings and manuals. Identify what software programs are utilized to generate these data products and what formats are available for delivered items.

6. Describe your company’s capability to obtain or manufacture molds/tooling. Also, provide information on your type of molding process.

7. What is your company’s current maximum production capacity per month? Provide information on any facility reserves you may possess to increase production capacity in the event of an immediate need do to critical operational mission requirements.

8. Are there specific requirements in the documentation that we provide that would currently preclude your product from being a viable solution to our requirement?

9. Include a detailed listing of facilities and equipment, including quantities, required to satisfy the requirements of this effort.
    a. Provide an outline of the proposed process, including inspections.

B. Hardware Production Questions:

This requirement involves fabrication and integration of a precision pressure and vacuum regulation system capable of maintaining a defined setpoint (27.7 ± 0.3 in H_2O) for a specified dwell period prior to measurement. Vendors must demonstrate experience in manufacturing and assembling calibration-sensitive systems that operate in a depot-level maintenance environment.

1. Describe your capability and experience in the manufacturing/fabrication of components.

C. Software Development Questions:

The system requires integrated control logic to regulate and maintain defined pressure and vacuum setpoints within tolerance prior to measurement. Software shall support closed-loop regulation, dwell timing, measurement capture, data display, and configuration control. Vendors should describe their software development practices, validation processes, and long-term support capability.

1. Is your company's software development processes certified to the Carnegie Mellon Capability Maturity Model? If so what level of certification have you achieved and what is the date of your certification.

2. Describe your capabilities and experience in managing software development projects, including subcontractor involvement. Include any experience in project planning, work breakdown structures, resource allocations, schedule tracking, risk analysis and cost management.

D. Repair Questions:

The system will operate in a depot maintenance environment and must be supportable long-term. Vendors shall demonstrate capability to provide technical manuals, calibration procedures, recommended spare parts lists, and sustainment support for a minimum of five years.

1. Describe your capabilities and experience in modifying existing systems/equipment (hardware and software) to solve maintenance and support problems in the depot environment.

2. Describe your capabilities and experience in developing/modifying repair or maintenance procedures. Include associated upgrade of technical orders and preparation of new technical orders.

3. Describe your configuration management processes and how you identify and resolve parts obsolescence and diminishing manufacturing sources problems?

4. Describe your process for maintaining inventory records and reporting on hand/in work balances and repair status to your customer.

E. Commerciality Questions:

1. Are there established catalog or market prices for our requirement? If you offer this product and/or service to both U.S. Government and commercial sources, is the same workforce used for both the U.S. Government and general public? Is our requirement offered to both under similar terms and conditions? Briefly describe any differences.

2. Describe your standard warranty and return process for goods and services furnished to the government for items similar in nature to this requirement.